

POLITECNICO
MILANO 1863

Temperature sensors
Part 2

SENSOR SYSTEMS

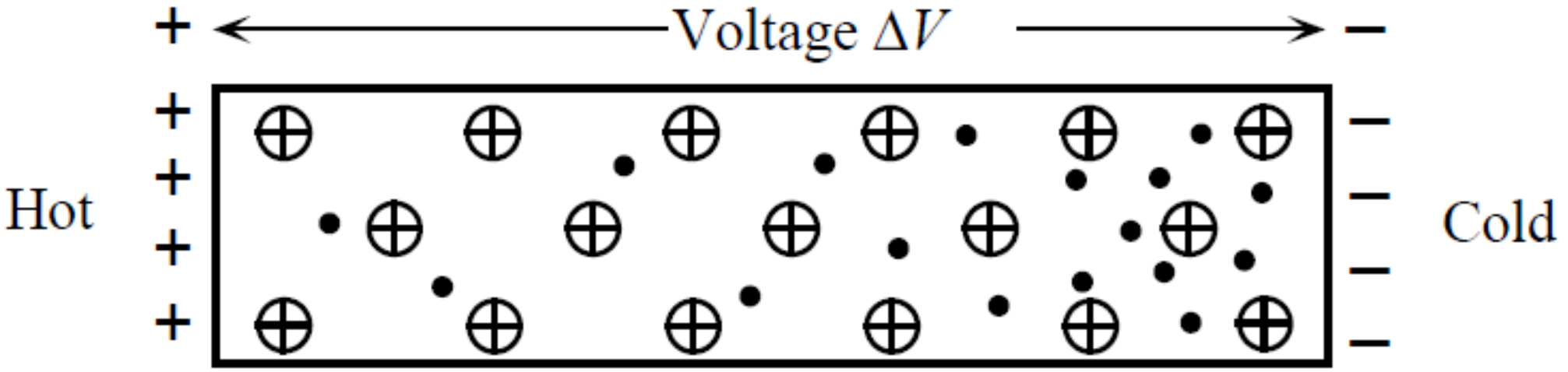
Federica Villa
federica.villa@polimi.it

- Resistance Temperature Detectors
- Thermistors
- Thermocouple
- Diode and bandgap temperature sensors
- Infrared thermometer



WeBeep “Supplementary materials”: 08 – Thermocouples

A temperature difference between two points in a conductor or semiconductor results in a voltage difference between these two points.



The electrons in the hot region are more energetic (higher energy levels)

→ net diffusion of electrons from hot to cold

→ voltage

(Mott formula)

= sensitivity of the conductor/semiconductor: $S(T) = \frac{dV}{dT}$

The sign represents the potential of the cold side with respect to the hot side.

→ Typically metals and n-doped semiconductors have negative *S*, whereas p-doped semiconductors have positive *S*.

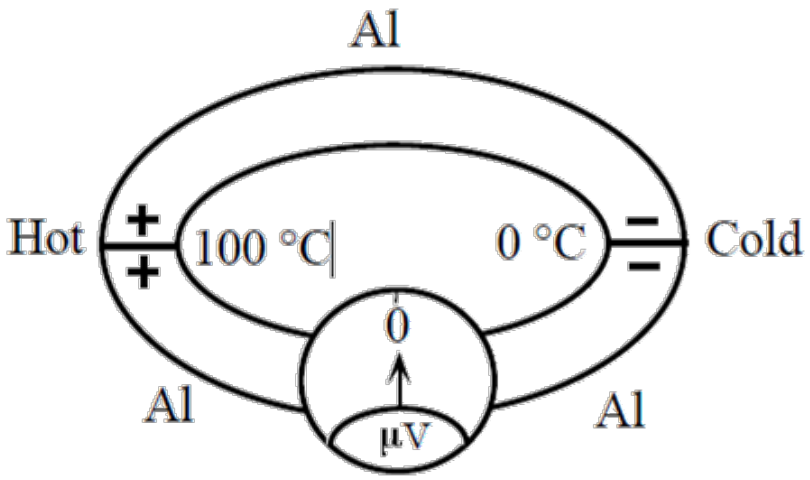
S depends on temperature, thus:

$$\Delta V = \int_{T_0}^{T} S(T) \cdot dT$$

Metal	*S* at 0°C ($\mu V\ K^{-1}$)	*S* at 27°C ($\mu V\ K^{-1}$)
Al	-1.60	-1.8
Pt	-4.45	-5.28
Constantan	-39.90	-35.00

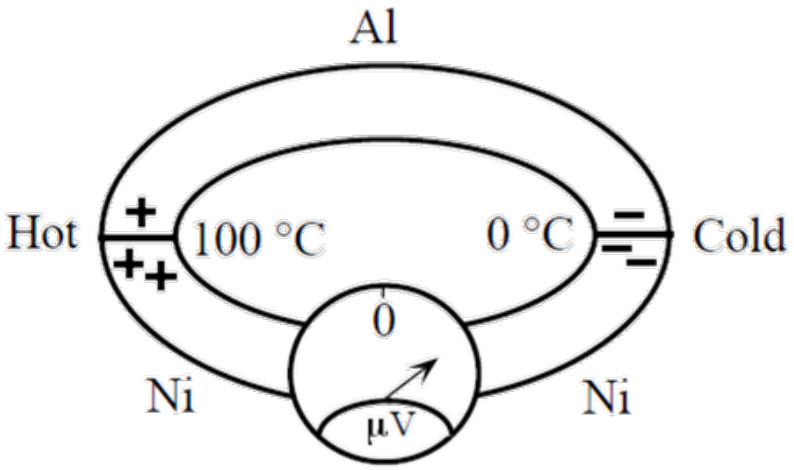
Only the net Seebeck voltage difference

between different metals can be measured.



If one metal is used

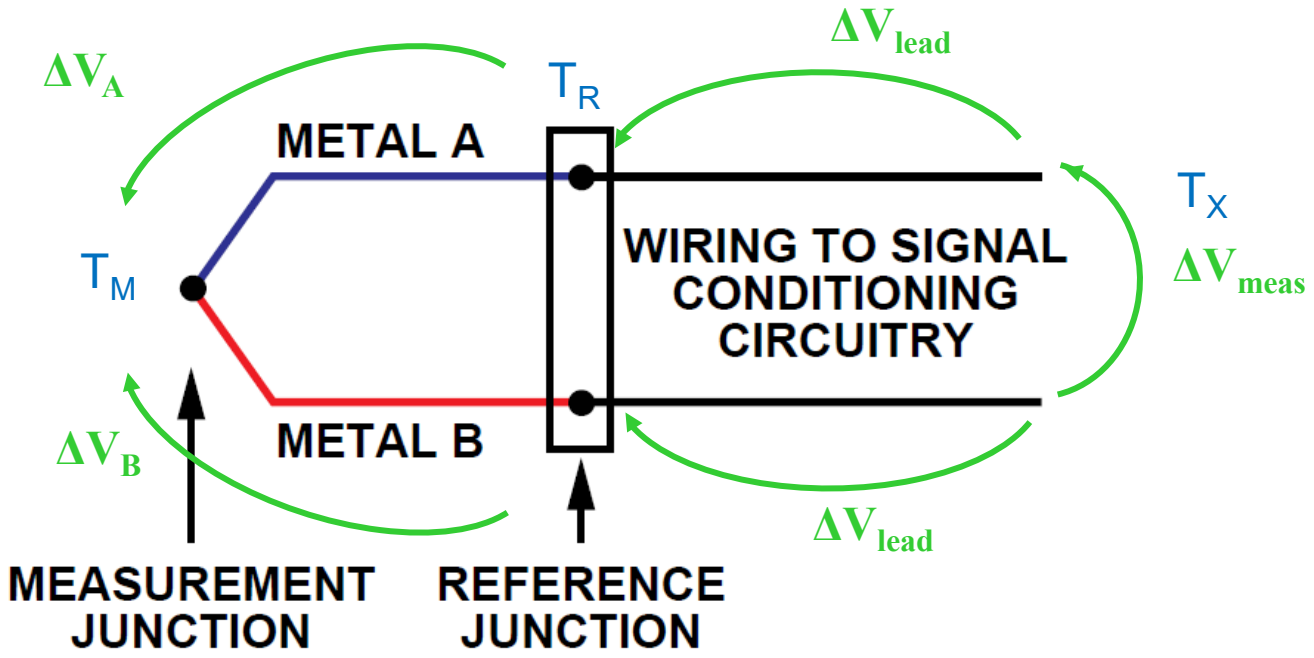
the net voltage is 0V.



If two metals with different

Seebeck coefficient are used

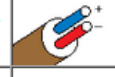
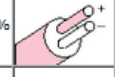
there is a net voltage.

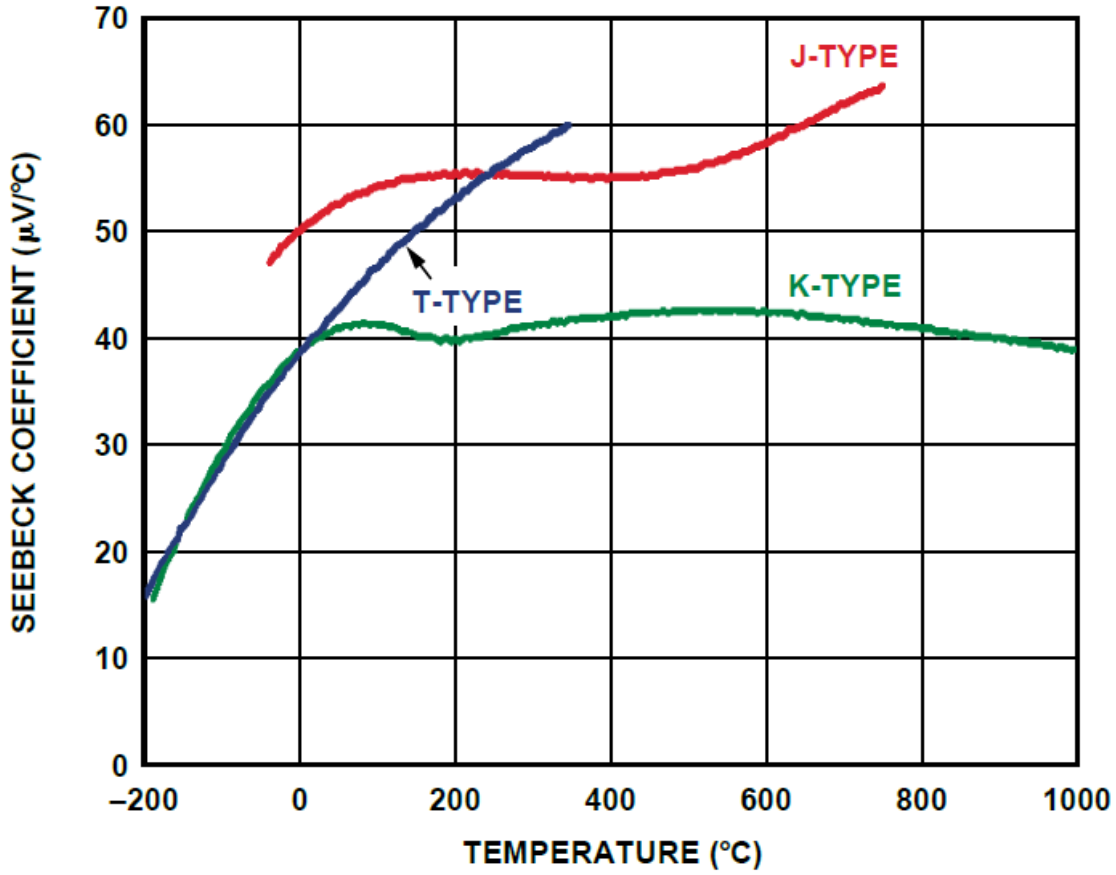


$$\Delta V_{meas} = \Delta V_{lead} + \Delta V_B - \Delta V_A - \Delta V_{lead} = \Delta V_B - \Delta V_A \quad \rightarrow \quad \Delta V_{meas} = \int_{T_R}^{T_M} (S_B(T) - S_A(T)) \cdot dT = \int_{T_R}^{T_M} S_{AB}(T) \cdot dT$$

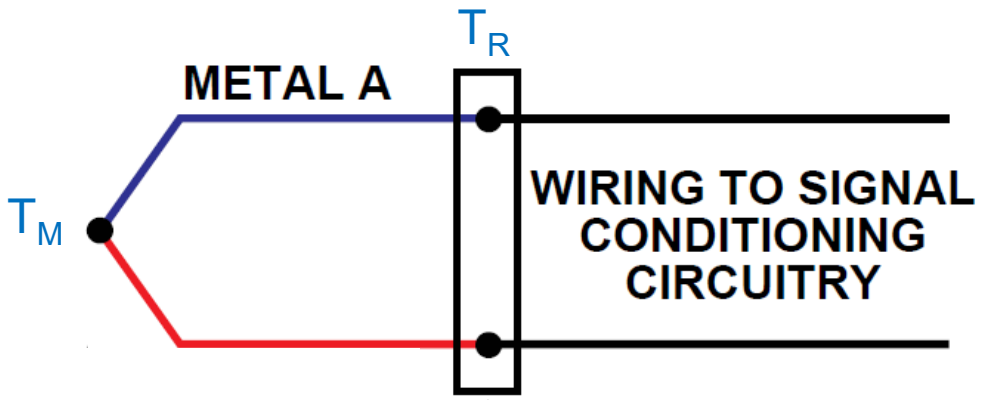
$$S_{AB}(T) = S_B(T) - S_A(T)$$

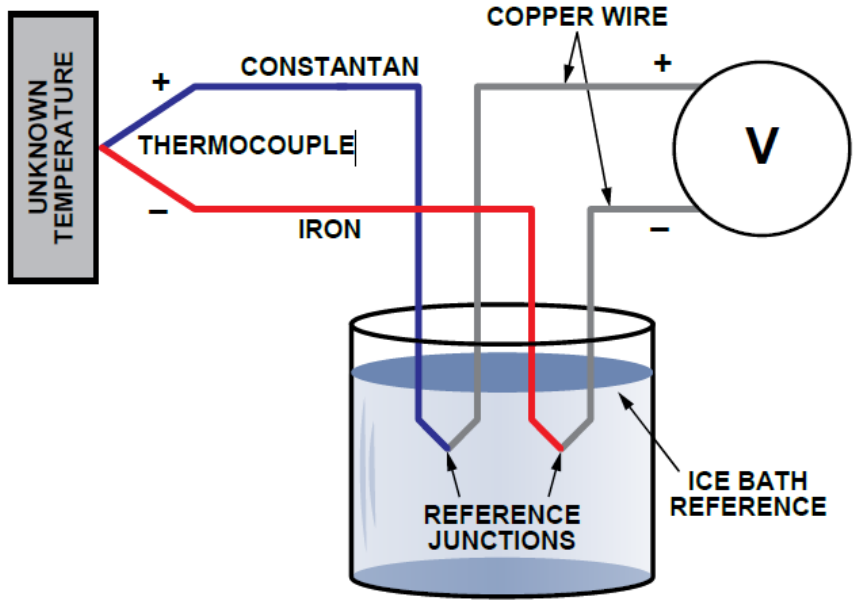
Thermocouples types

ANSI Code	Alloy Combination		Color Coding		Maximum Useful Temperature Range ++	Maximum Thermocouple Grade Temperature Range	EMF (mV) Over Max. Temperature Range	Standard Limits of Error** (above 0°C)	Special Limits of Error** (above 0°C)	International IEC 584-3	Comments Environment — Bare Wire	IEC Code
	+ Lead	- Lead	Thermocouple Grade	Extension Grade								
J	IRON Fe (magnetic)	CONSTANTAN COPPER-NICKEL Cu-Ni			Thermocouple Grade: 32 to 1382°F 0 to 750°C Extension Grade: 32 to 392°F 0 to 200°C	-346 to 2193°F -210 to 1200°C	-8.095 to 69.553	greater of 2.2°C or 0.75%	greater of 1.1°C or 0.4%		Reducing, Vacuum, Inert. Limited Use in Oxidizing at High Temperatures. Not Recommended for Low Temperatures.	J
K	CHROMEGA® NICKELCHROMIUM Ni-Cr	ALOMEGA® NICKEL-ALUMINUM Ni-AI (magnetic)			Thermocouple Grade: -328 to 2282°F -200 to 1250°C Extension Grade: 32 to 392°F 0 to 200°C	-454 to 2501°F -270 to 1372°C	-6.458 to 54.886	greater of 2.2°C or 0.75%	greater of 1.1°C or 0.4%		Clean Oxidizing and Inert. Limited Use in Vacuum or Reducing. Wide Temperature Range, Most Popular Calibration	K
T	COPPER Cu	CONSTANTAN COPPER-NICKEL Cu-Ni			Thermocouple Grade: -328 to 662°F -250 to 350°C Extension Grade: -76 to 212°F -60 to 100°C	-454 to 752°F -270 to 400°C	-6.258 to 20.872	greater of 1.0°C or 0.75%	greater of 0.5°C or 0.4%		Mild Oxidizing, Reducing Vacuum or Inert. Good Where Moisture Is Present. Low Temperature and Cryogenic Applications	T
E	CHROMEGA® NICKELCHROMIUM Ni-Cr	CONSTANTAN COPPER-NICKEL Cu-Ni			Thermocouple Grade: -328 to 1652°F -200 to 900°C Extension Grade: 32 to 392°F 0 to 200°C	-454 to 1832°F -270 to 1000°C	-9.835 to 76.373	greater of 1.7°C or 0.5%	greater of 1.0°C or 0.4%		Oxidizing or Inert. Limited Use in Vacuum or Reducing. Highest EMF Change Per Degree	E
N	OMEGA-P® NICROSIL Ni-Cr-Si	OMEGA-N® NISIL Ni-Si-Mg			Thermocouple Grade: -450 to 2372°F -270 to 1300°C Extension Grade: 32 to 392°F 0 to 200°C	-450 to 2372°F -270 to 1300°C	-4.345 to 47.513	greater of 2.2°C or 0.75%	greater of 1.1°C or 0.4%		Alternative to Type K. More Stable at High Temps	N
R	PLATINUM- 13% RHODIUM Pt-13% Rh	PLATINUM Pt	NONE ESTABLISHED		Thermocouple Grade: 32 to 2642°F 0 to 1450°C Extension Grade: 32 to 300°F 0 to 150°C	-58 to 3214°F -50 to 1768°C	-0.226 to 21.101	greater of 1.5°C or 0.25%	greater of 0.6°C or 0.1%		Oxidizing or Inert. Do Not Insert in Metal Tubes. Beware of Contamination. High Temperature	R
S	PLATINUM- 10% RHODIUM Pt-10% Rh	PLATINUM Pt	NONE ESTABLISHED		Thermocouple Grade: 32 to 2642°F 0 to 1400°C Extension Grade: 32 to 300°F 0 to 150°C	-58 to 3214°F -50 to 1768°C	-0.236 to 18.693	greater of 1.5°C or 0.25%	greater of 0.6°C or 0.1%		Oxidizing or Inert. Do Not Insert in Metal Tubes. Beware of Contamination. High Temperature	S
U	COPPER Cu	COPPER-LOW NICKEL Cu-Ni	NONE ESTABLISHED		Extension Grade: 32 to 122°F 0 to 50°C						Extension Grade Connecting Wire for R and S Thermocouples, Also Known as RX and SX Extension Wire.	U
B	PLATINUM- 30% RHODIUM Pt-30% Rh	PLATINUM- 6% RHODIUM Pt-6% Rh	NONE ESTABLISHED		Thermocouple Grade: 32 to 3092°F 0 to 1700°C Extension Grade: 32 to 212°F 0 to 100°C	32 to 3308°F 0 to 1820°C	0 to 13.820	0.5% over 800°C	NOT ESTABLISHED		Oxidizing or Inert. Do Not Insert in Metal Tubes. Beware of Contamination. High Temperature. Common Use in Glass Industry	B
G* (W)	TUNGSTEN W	TUNGSTEN- 26% RHENIUM W-26% Re	NONE ESTABLISHED		Thermocouple Grade: 32 to 4208°F 0 to 2320°C Extension Grade: 32 to 500°F 0 to 260°C	32 to 4208°F 0 to 2320°C	0 to 38.564	greater of 4.5°C or 1.0%	NOT ESTABLISHED		Vacuum, Inert, Hydrogen. Beware of Embrittlement. Not Practical Below 399°C (750°F). Not for Oxidizing Atmosphere	G (W)
C* (W5)	TUNGSTEN- 5% RHENIUM W-5% Re	TUNGSTEN- 26% RHENIUM W-26% Re	NONE ESTABLISHED		Thermocouple Grade: 32 to 4208°F 0 to 2320°C Extension Grade: 32 to 1600°F 0 to 870°C	32 to 4208°F 0 to 2320°C	0 to 37.066	greater of 4.5°C or 1.0%	NOT ESTABLISHED		Vacuum, Inert, Hydrogen. Beware of Embrittlement. Not Practical Below 399°C (750°F) Not for Oxidizing Atmosphere	C (W5)
D* (W3)	TUNGSTEN- 3% RHENIUM W-3% Re	TUNGSTEN- 25% RHENIUM W-25% Re	NONE ESTABLISHED		Thermocouple Grade: 32 to 4208°F 0 to 2320°C Extension Grade: 32 to 500°F 0 to 260°C	32 to 4208°F 0 to 2320°C	0 to 39.506	greater of 4.5°C or 1.0%	NOT ESTABLISHED		Vacuum, Inert, Hydrogen. Beware of Embrittlement. Not Practical Below 399°C (750°F)–Not for Oxidizing Atmosphere	D (W3)



→ The voltage signal is not linear; a calibration is required.





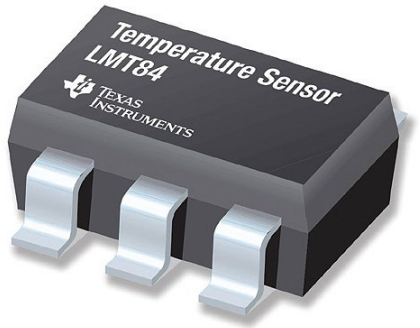
NOT PRACTICAL !!

Only relative temperatures

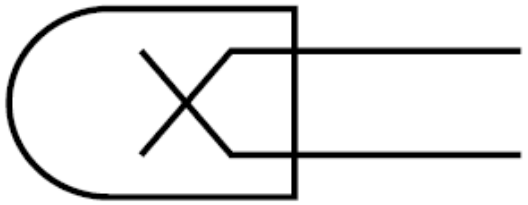
Reference-junction compensation

= the reference junction temperature is measured with another temperature sensitive device:

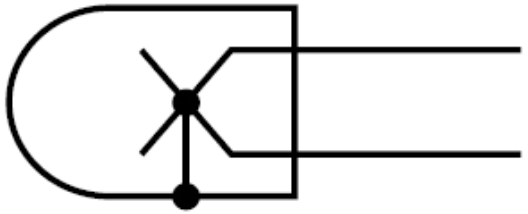
- RDT
- thermistor
- thermal diode
- integrated temperature sensor



Grounding options



INSULATED

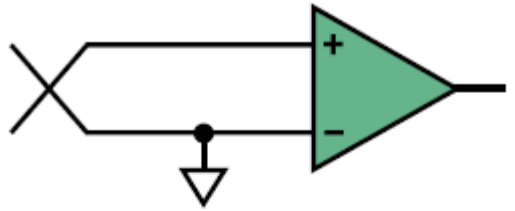


GROUNDED

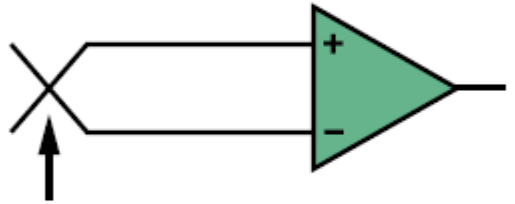


EXPOSED

WHEN USING ISOLATED THERMOCOUPLE TIPS

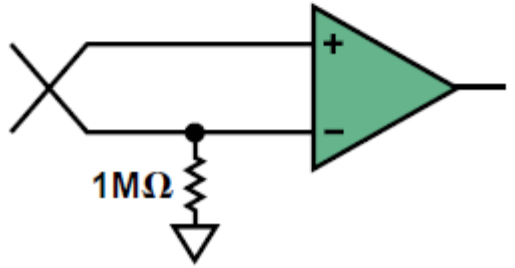


WHEN USING EXPOSED OR GROUNDED THERMOCOUPLE TIPS

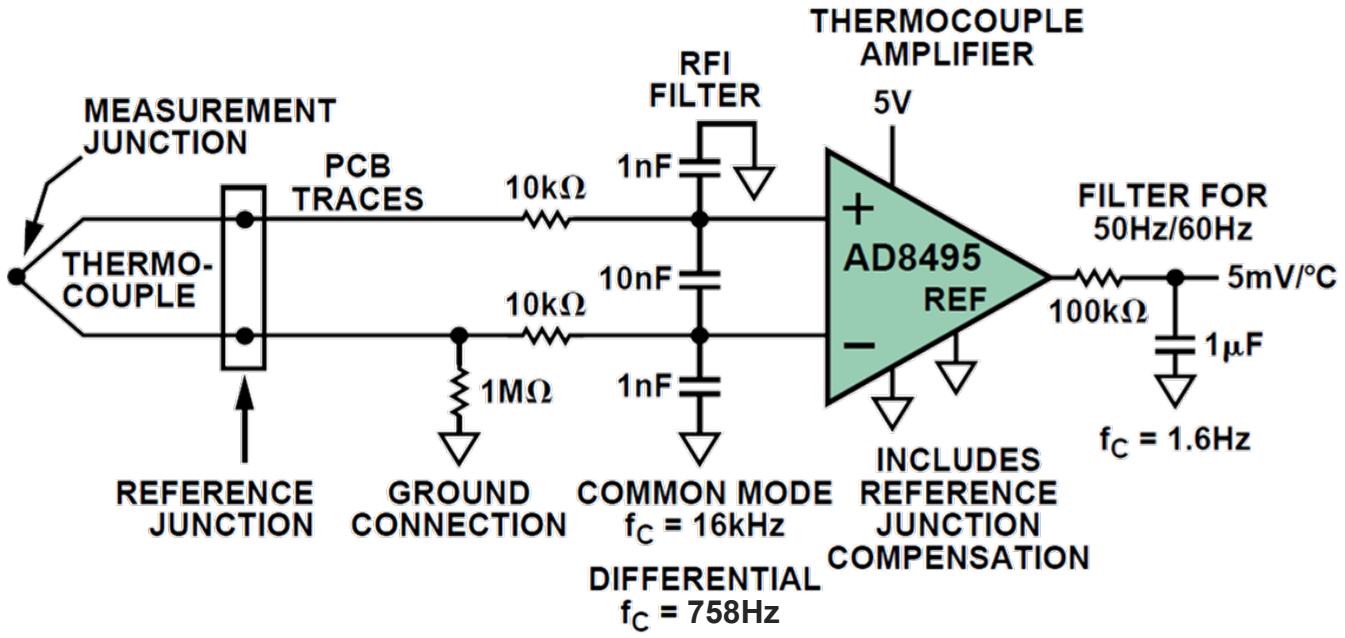


ELECTRICAL CONNECTION OCCURS AT TIP. VOLTAGE MUST STAY IN COMMON-MODE INPUT RANGE OF AMPLIFIER

WHEN THERMOCOUPLE TIP TYPE IS UNKNOWN



Measuring circuits examples (1/2)



INA

Gain = 122

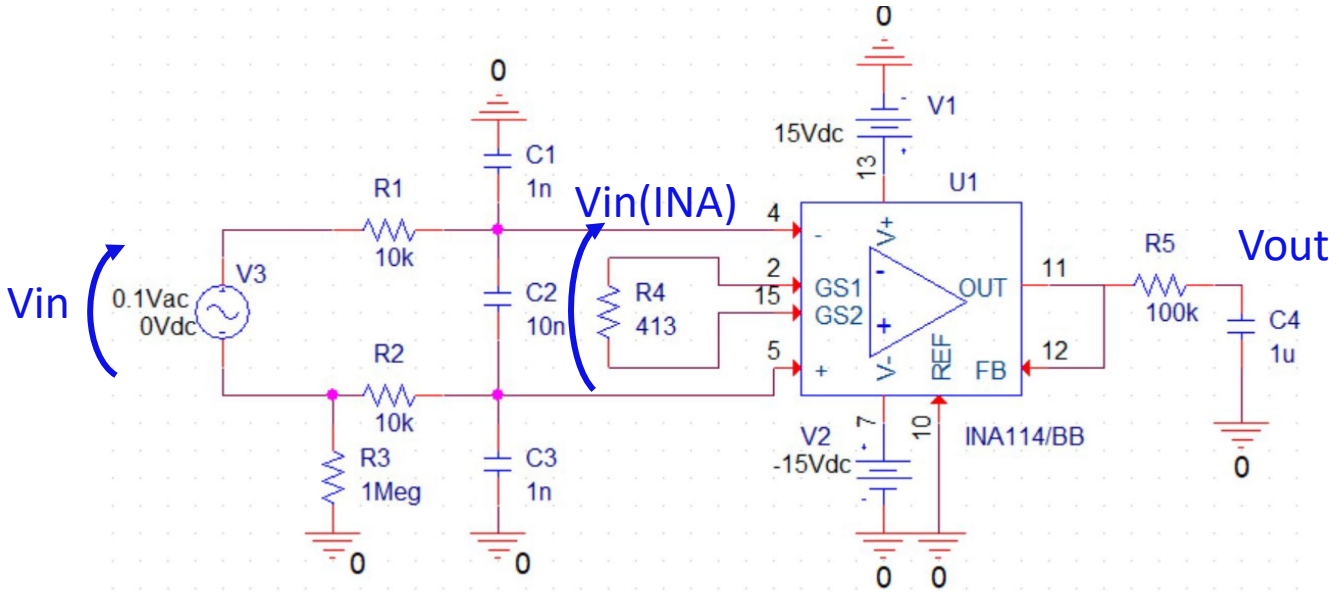
Ref Temperature compensation

Non-linearity correction for k-thermocouples

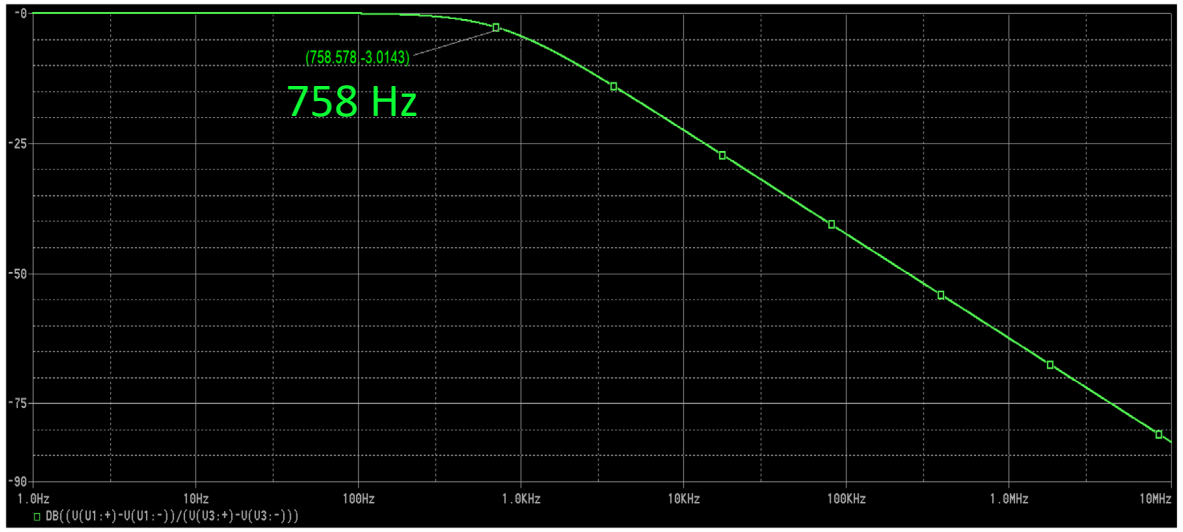
Noise/disturbances reduction:

- Differential input (eliminates common mode disturbances)
- Radio Frequency Interference filter (low pass filter)
- Filter for 50/60Hz

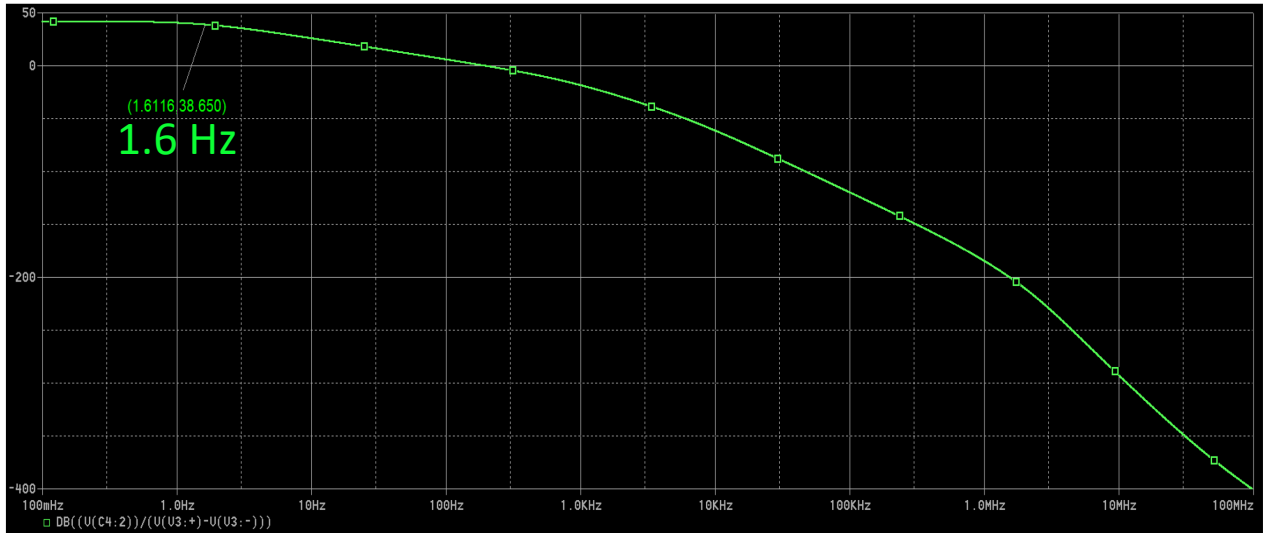
Differential mode



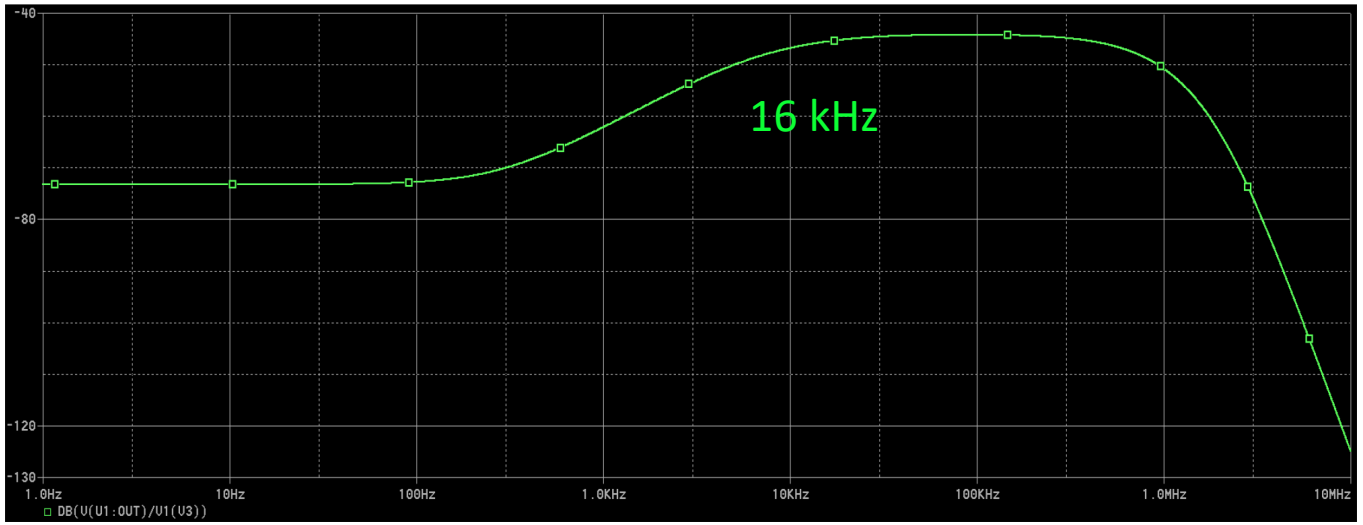
Vin(INA) vs Vin



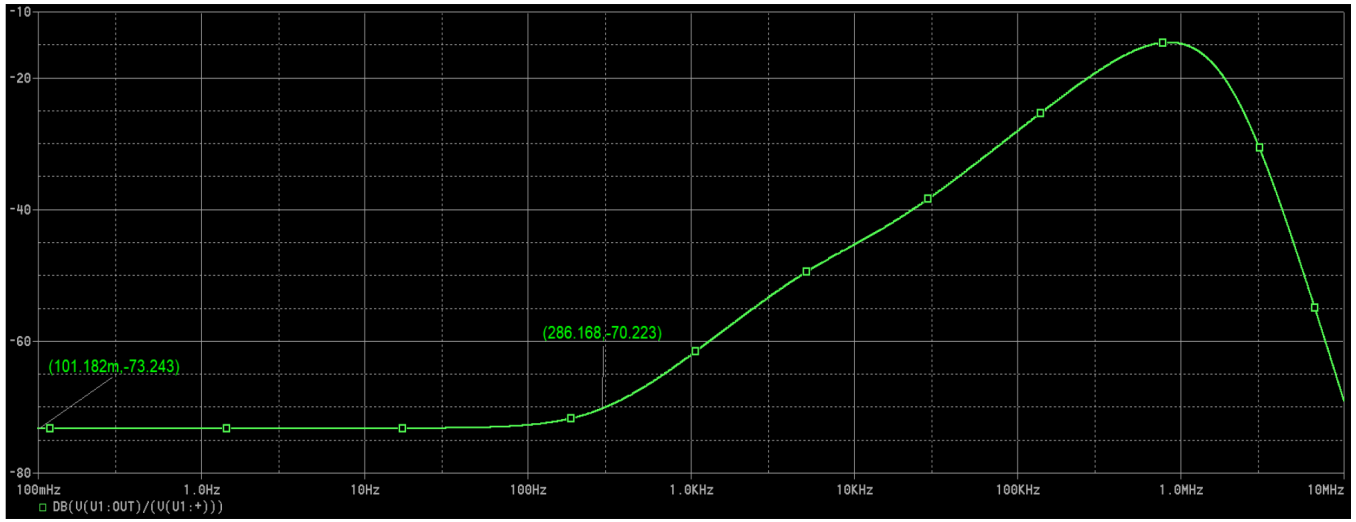
Vout vs Vin

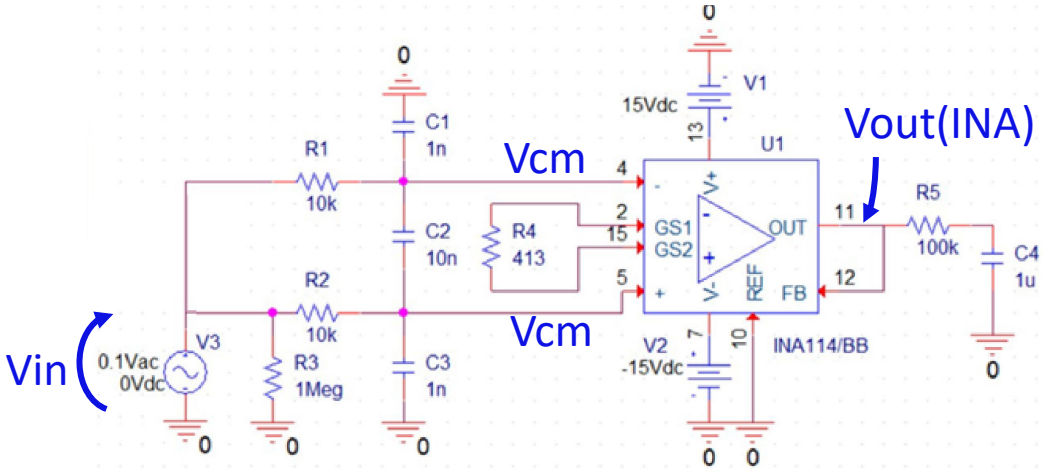


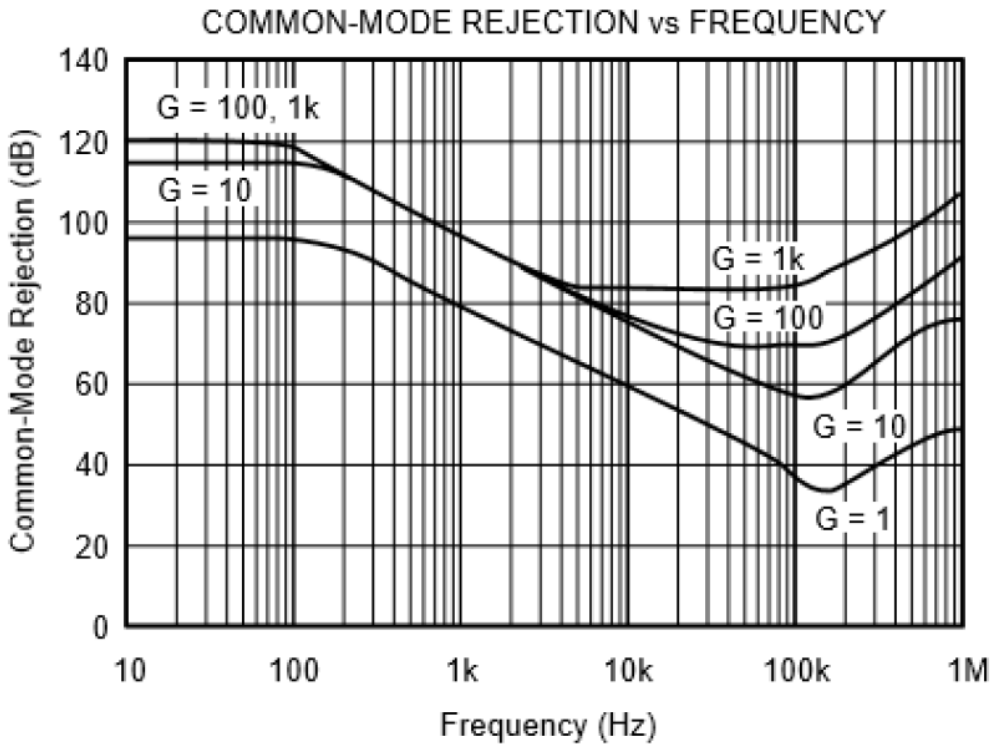
Vout(INA) vs Vin



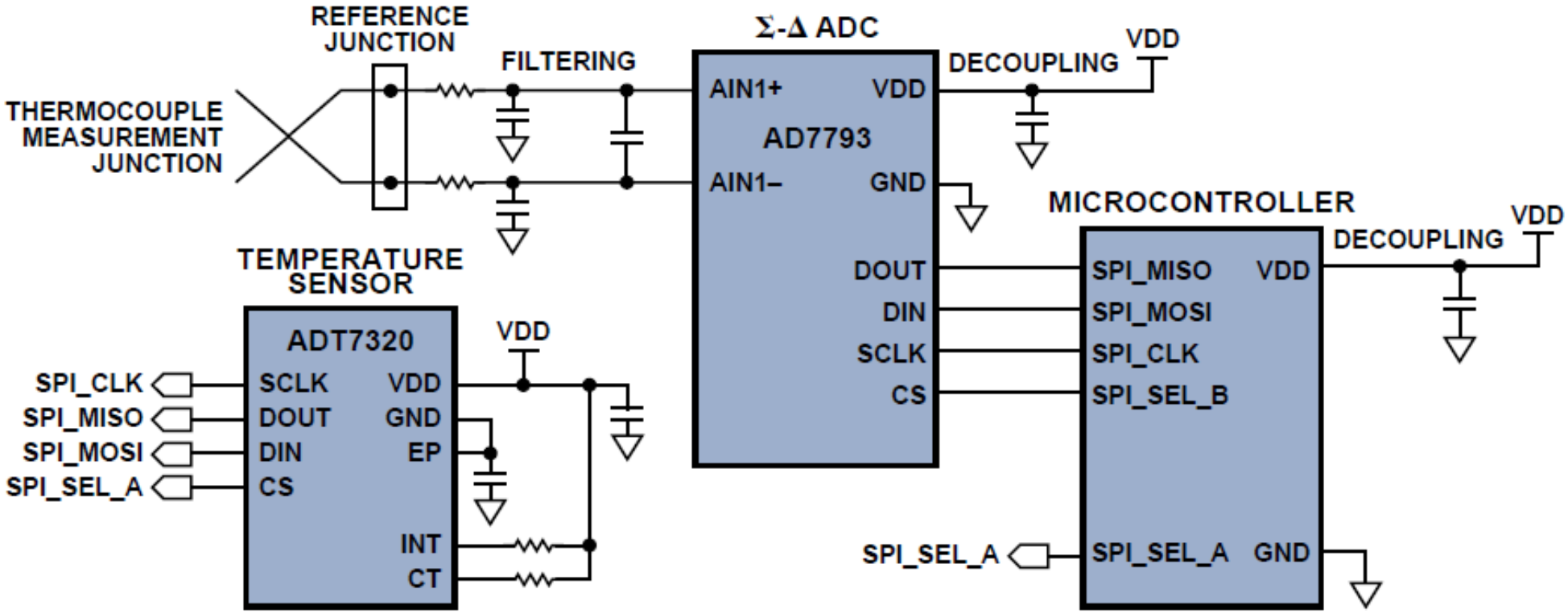
Vout(INA) vs Vcm(INA)







Measuring circuits examples (2/2)



- Flexible (J-, K-, T-type thermocouples)
- Optimized for accuracy
- INA included in the ADC

Thermocouple pros and cons

Pros

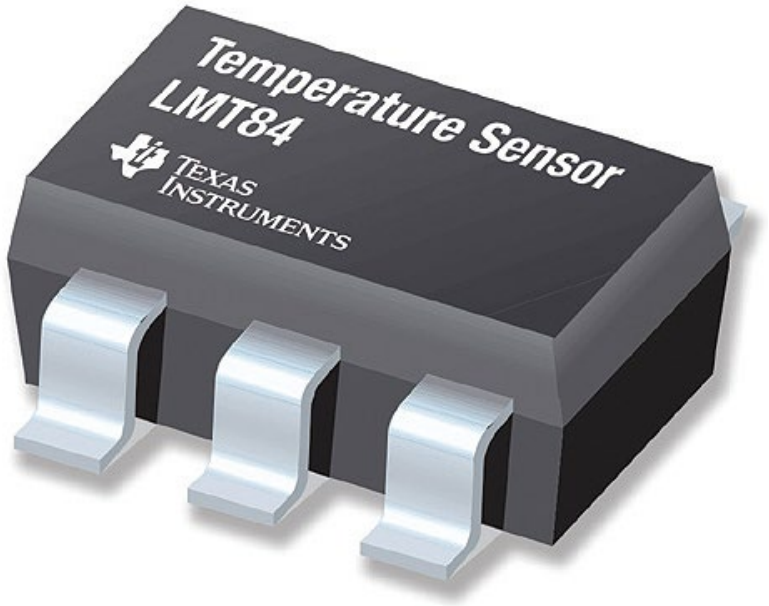
- Wide measuring ranges including very high limits
- Fast response times
- Tiny measuring point
- Moderate price
- No self heating
- Robust to mechanical stress

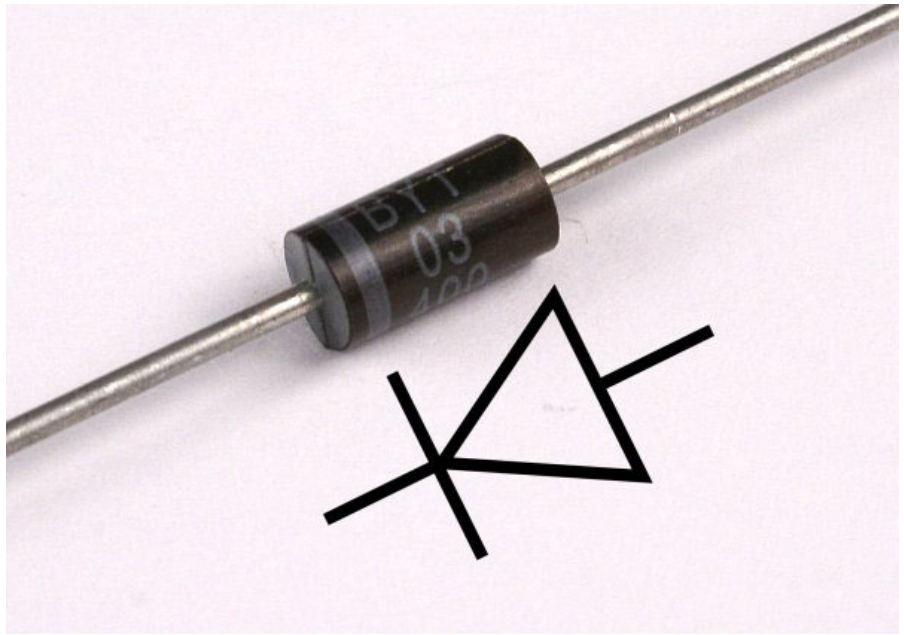
Cons

- Low sensitivity
- Signal strength is very low and prone to EMI problems
- Low linearity
- Only relative temperature (not absolute)

RTD, thermistor, termocouple comparison

	RTD	Thermistor	Thermocouple
sensitivity	0.4 Ω/°C	**200 Ω/°C**	40-50 µV/°C
range	-200°C to 800°C	-100°C to 500°C	**-240°C to 2400°C**
accuracy	**0.5%**	1%	0.75%
linearity	**Best**	Not linear	Good
cost	≈ 5-10€	**≈ 0.5€**	≈ 5-10€


Temperature Sensor
LMT84
Texas Instruments



$$I = I_s \cdot \left(e^{\frac{qV}{mkT}} - 1\right)$$

I_s reverse saturation current

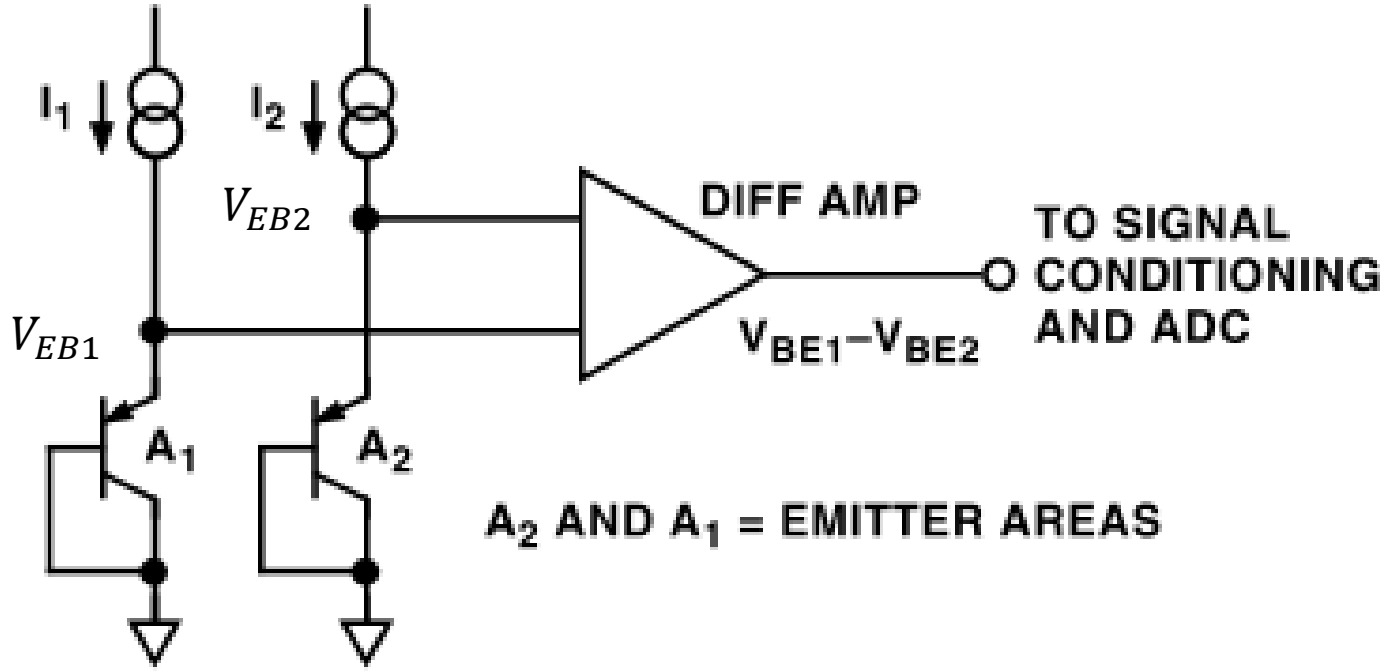
m technology parameter

$$V = m\frac{kT}{q} \cdot ln\left(\frac{I}{I_S} + 1\right)$$

$$\frac{\Delta V}{\Delta T} = m\frac{k}{q} \cdot ln\left(\frac{I}{I_S} + 1\right)$$

Not linear since I_s depends on temperature

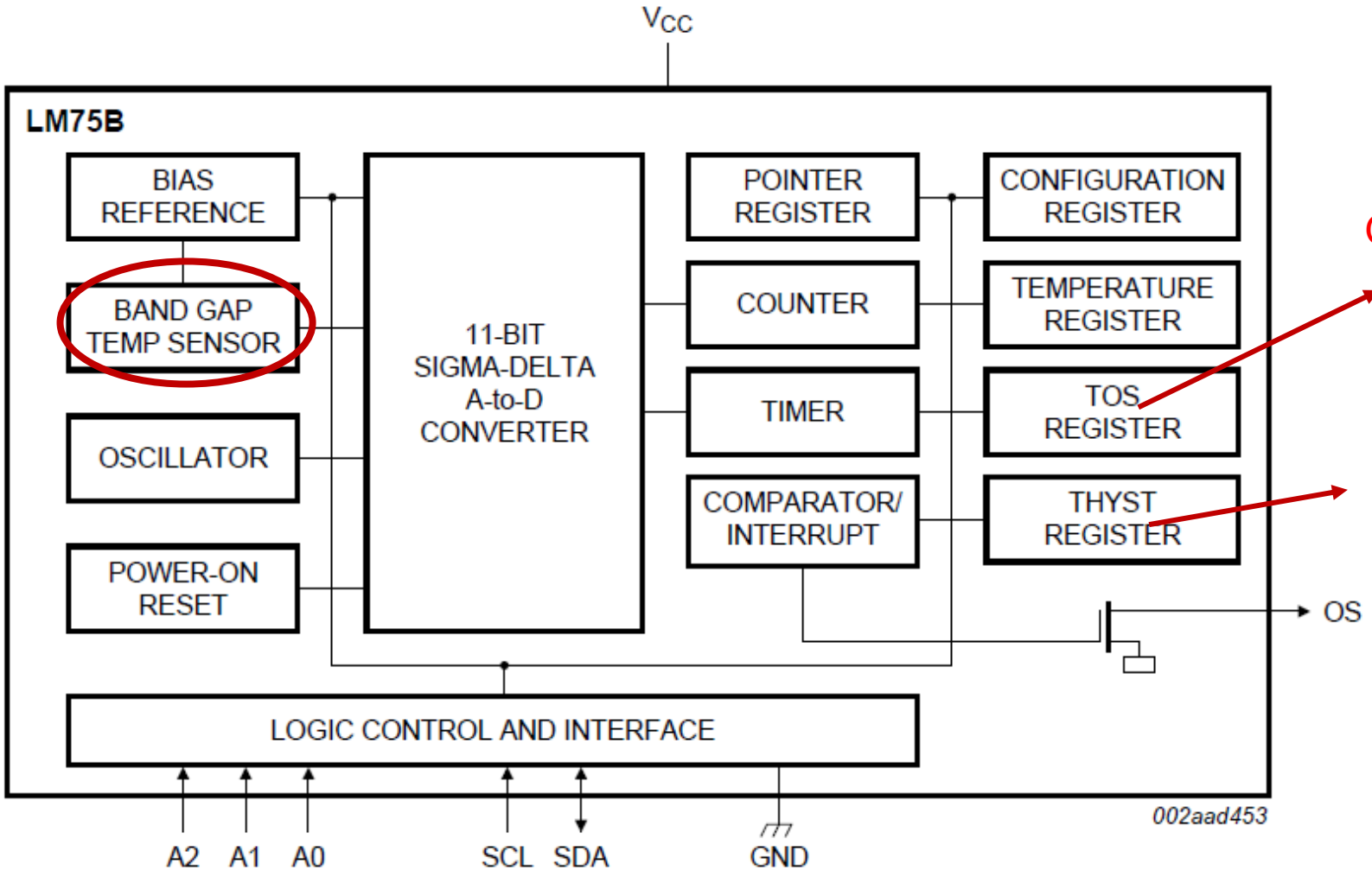
$$V = m\frac{kT}{q} \cdot ln\left(\frac{I}{I_S} + 1\right)$$



$$V_{EB1} \approx \frac{kT}{q} \cdot ln\left(\frac{I_1}{A_1 J_S}\right) \quad V_{EB2} \approx \frac{kT}{q} \cdot ln\left(\frac{I_2}{A_2 J_S}\right) \rightarrow V_{EB1} - V_{EB2} = \frac{kT}{q} \cdot ln\left(\frac{I_1 A_2}{I_2 A_1}\right)$$

If $I_1 = I_2$ and $\frac{A_2}{A_1} = r$ → $V_{EB1} - V_{EB2} = \frac{kT}{q} \cdot \ln(r)$

$T_{measured} = const \cdot (V_{EB1} - V_{EB2})$ with $const = \frac{q}{k \cdot \ln(r)}$



Over-temperature shutdown

Hysteresis temperature

I^2C interface

Bandgap temperature sensors pros and cons

Pros

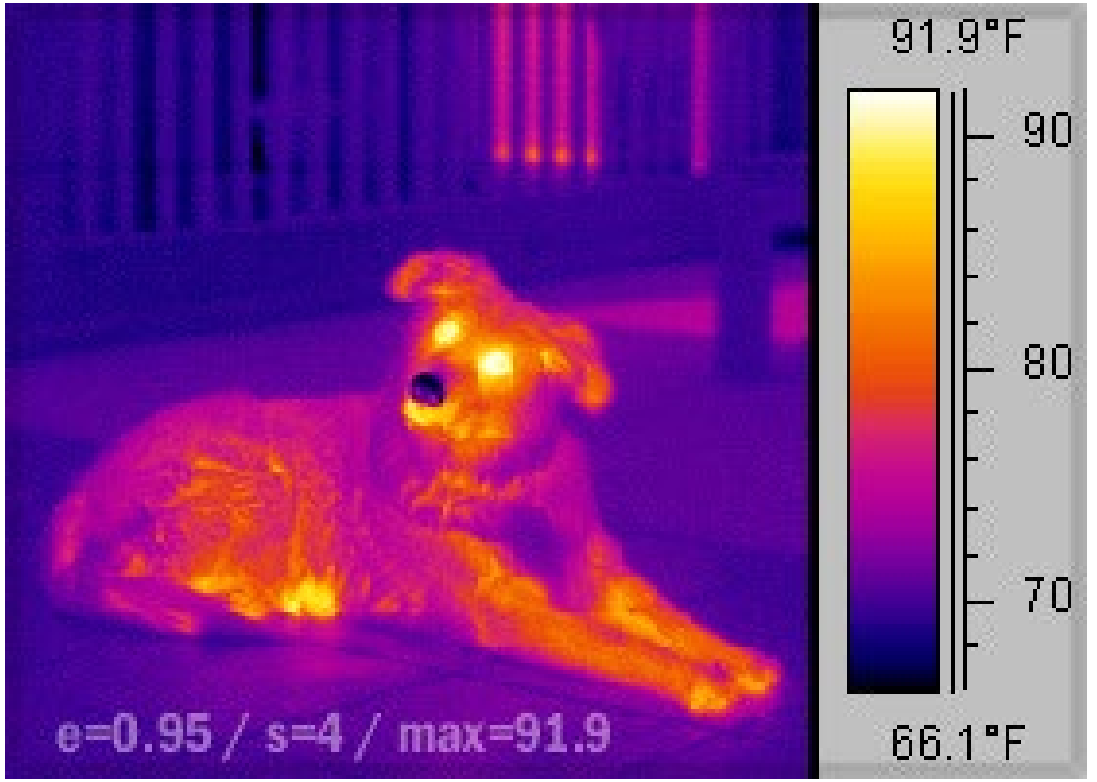
- Can be integrated in CMOS technology
- No need for external components
- Small size and low voltage
- High linearity
- Good accuracy

Cons

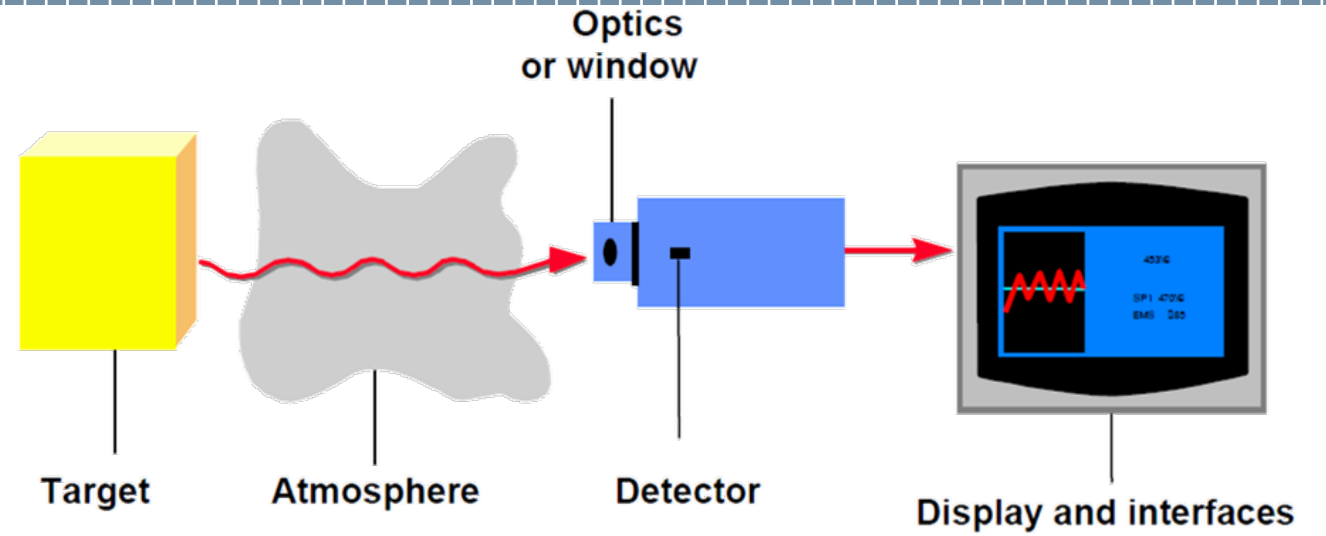
- Limited temperature range (-40°C - +125°C)

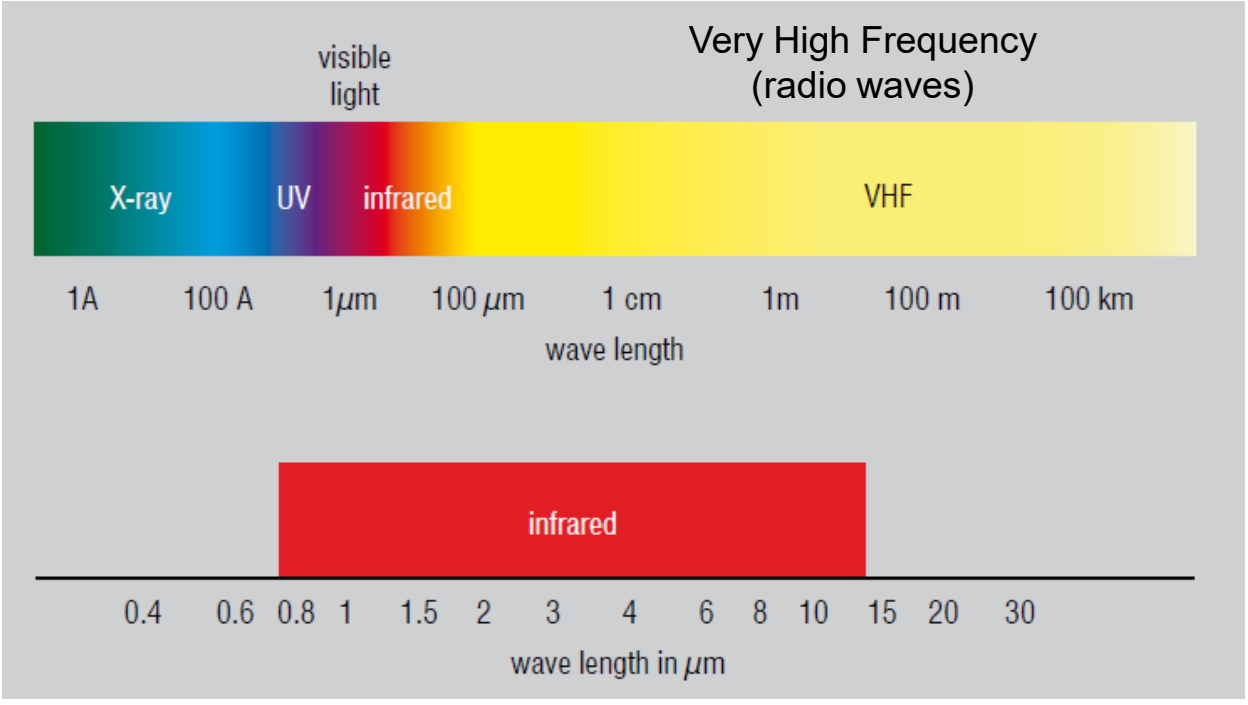
Infrared thermometer



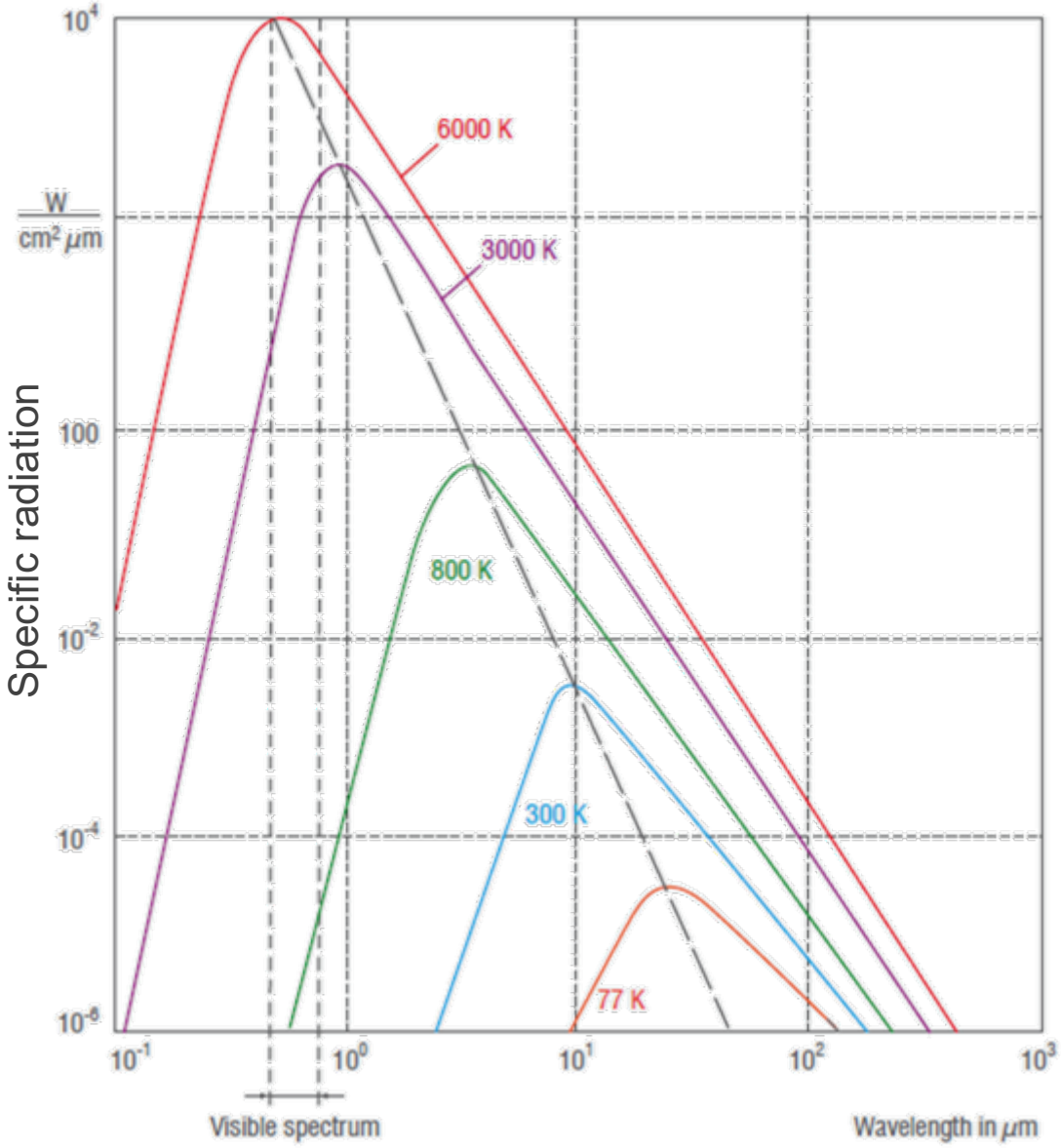


WeBeep “Supplementary materials”: 09 – Infrared thermometer


Optics
or window
Target
Atmosphere
Detector
Display and interfaces


visible
light
Very High Frequency
(radio waves)
X-ray
UV
infrared
VHF
1A
100 A
1μm
100 μm
1 cm
1m
100 m
100 km
wave length
infrared
0.4
0.6
0.8
1
1.5
2
3
4
6
8
10
15
20
30
wave length in μm

Black-body emission



Black-body:

A = ε = 1 R =T = 0

A absorption
ε emissivity
T transmissivity
R reflection

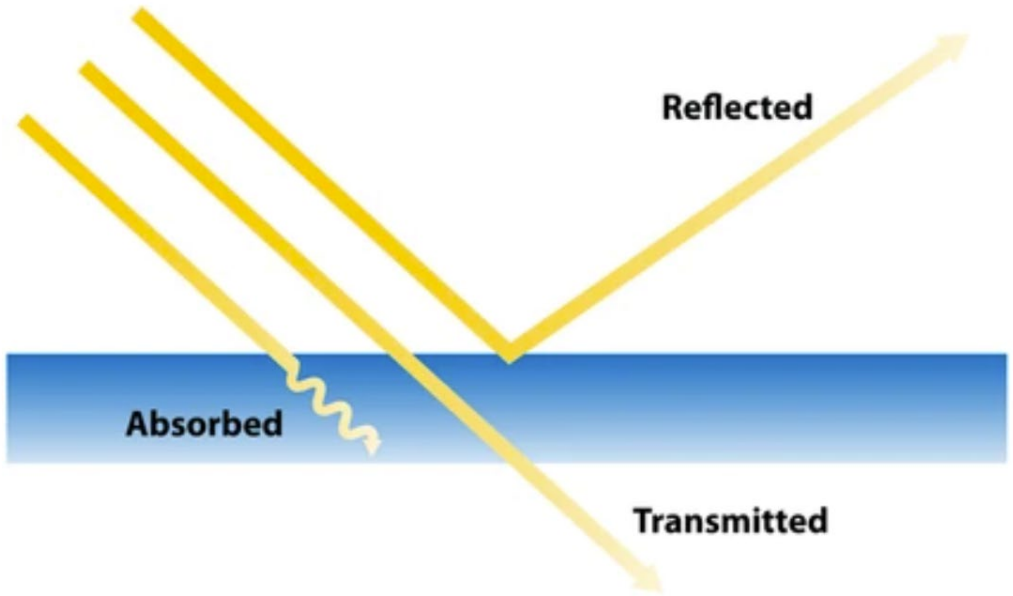
Stefan-Boltzmann law:

$$\frac{P}{A} = \varepsilon \cdot \sigma \cdot T^4$$

$\sigma = 5.67 \cdot 10^{-8}$ W/(m²K⁴)
Stefan-Boltzmann constant

Wien displacement law:

$\lambda_{max} \cdot T = 2898\ \mu m \cdot K$



Generic grey-body:
R + T + A = 1

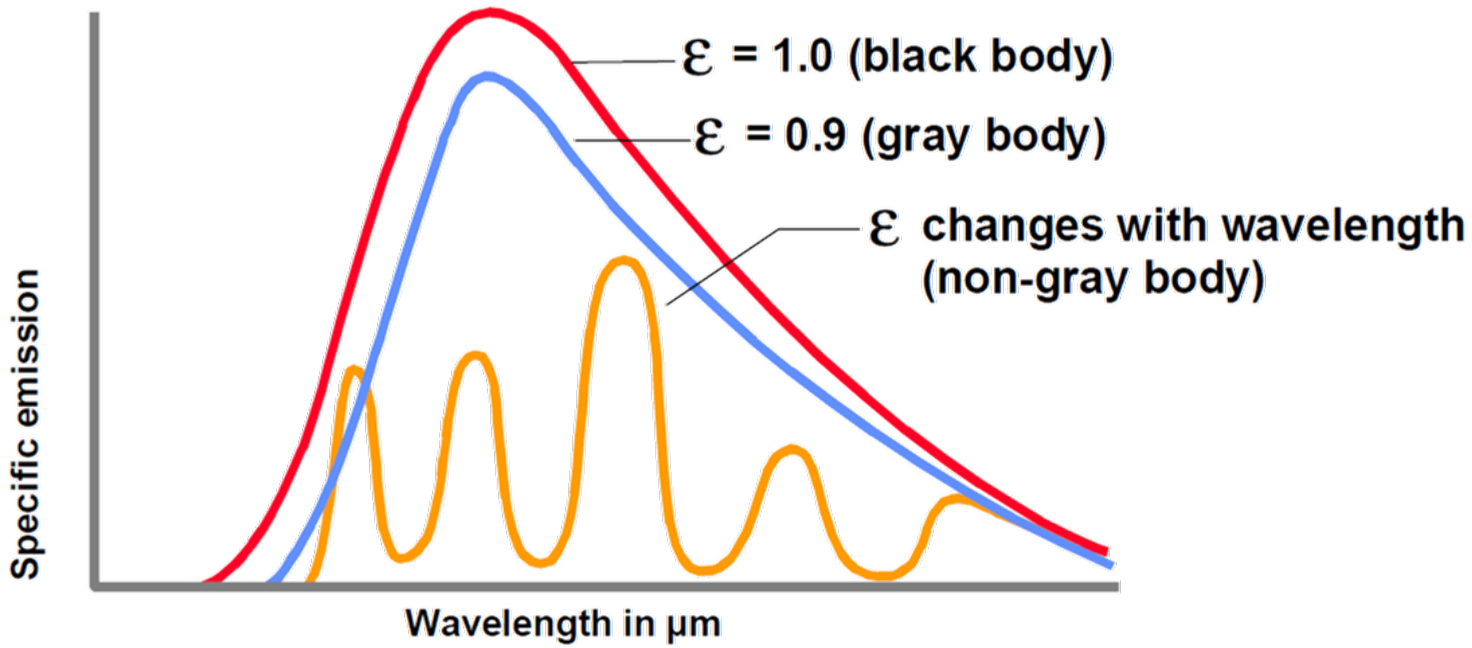
A = ε < 1

Solid grey-body:
R + A = 1 T = 0

A = ε < 1

Non-metallic materials (wood, plastic, rubber, organic materials, rock)
→ low reflectivity, high emissivity (0.8 - 0.95)
Metals (especially those with polished or shiny surfaces)
→ low emissivity (around 0.1)

→ Need of a calibration to estimate ε



Non gray-body: non-oxidized metals, glass, plastic films

→ Better to consider a specific wavelength

Determining emissivity

1) Determine temperature very accurately using a contact thermometer.
Then measure the target temperature with the IR thermometer and adjust ε.
Keep ε for all future measurements of that target.

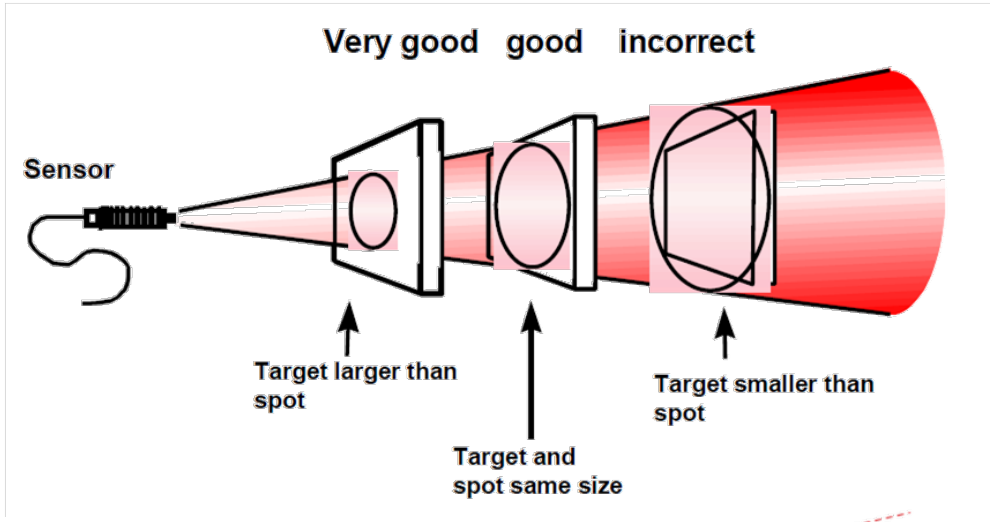




2) Attach to the target a plastic sticker with known ε and use the infrared sensor to determine the temperature with the known ε.



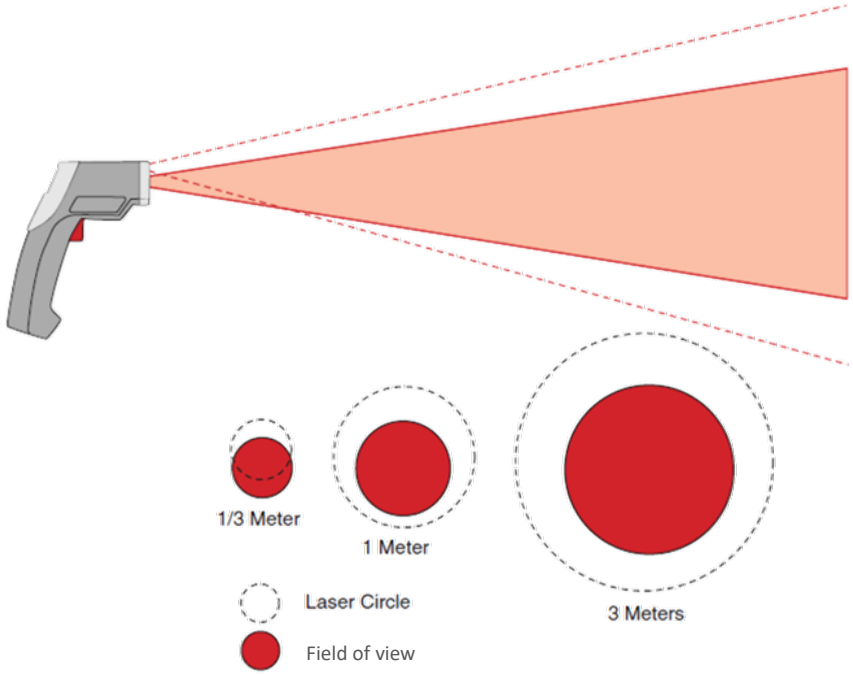
Measurement spot



Single spot sensor

or

cameras



Laser pointer

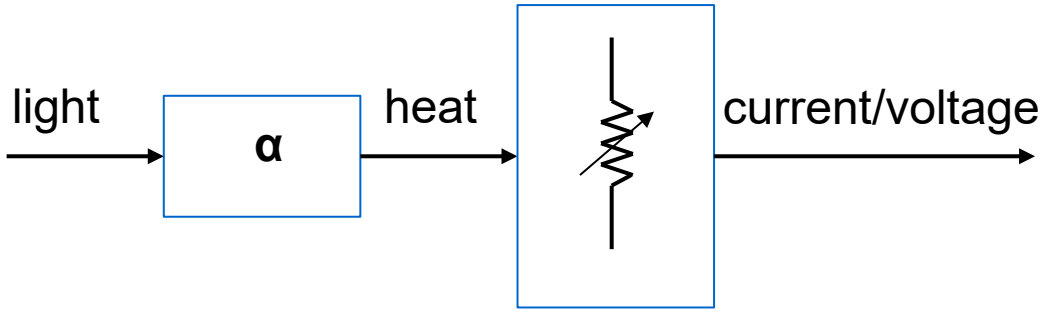
for exact spot size making

(even very precise coaxial pointers)

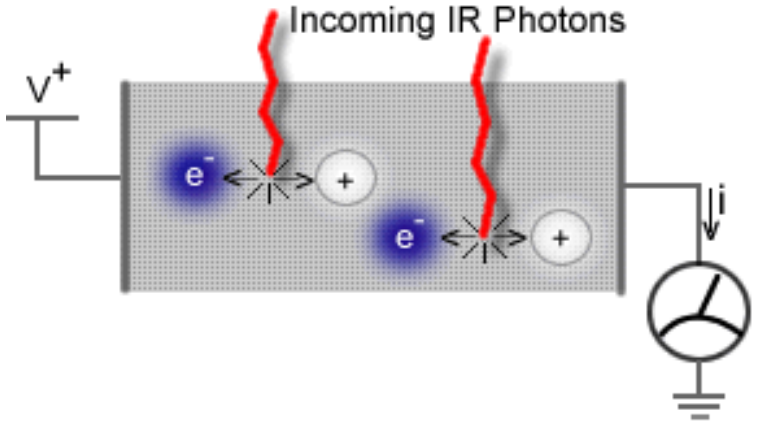
Infrared detectors

Thermal detectors:
- Thermopile detector
- Bolometer FPA (for IR cameras)
  (Focal Plane Array)
- Pyroelectric detector

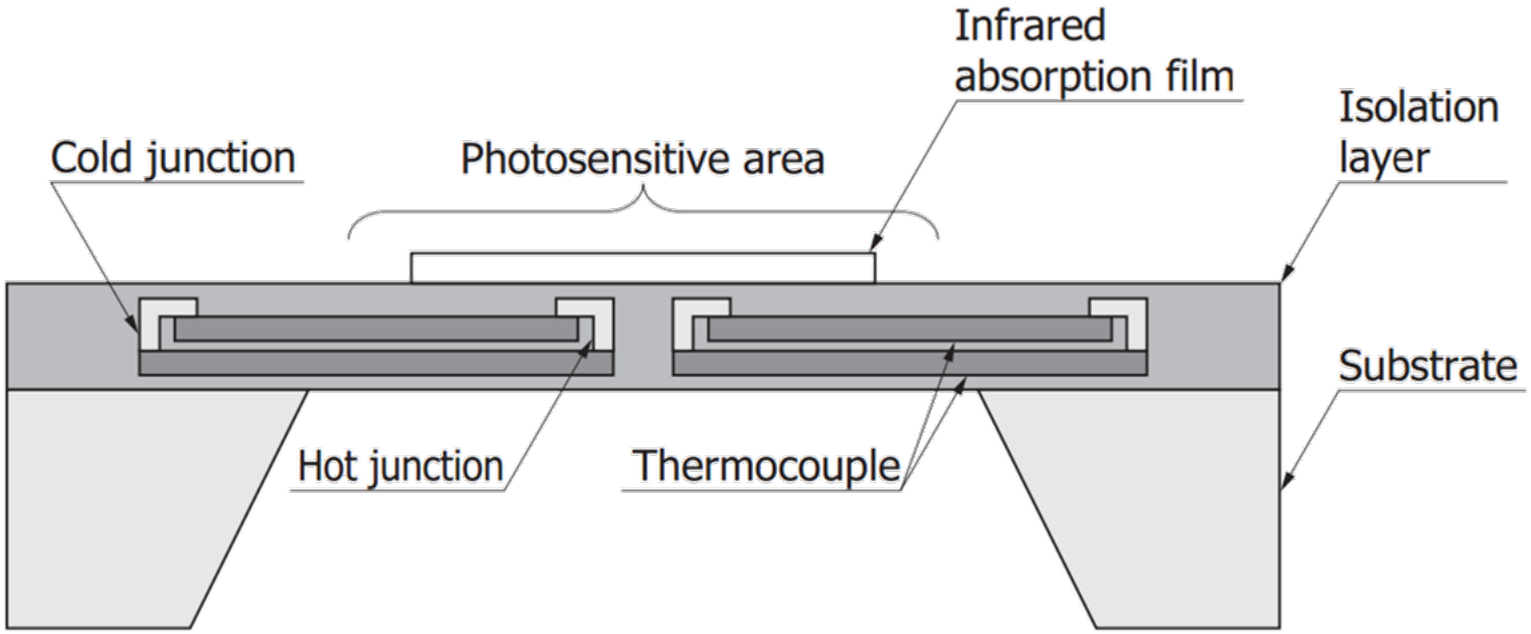
Quantum detectors



Thermal detector



Quantum detector

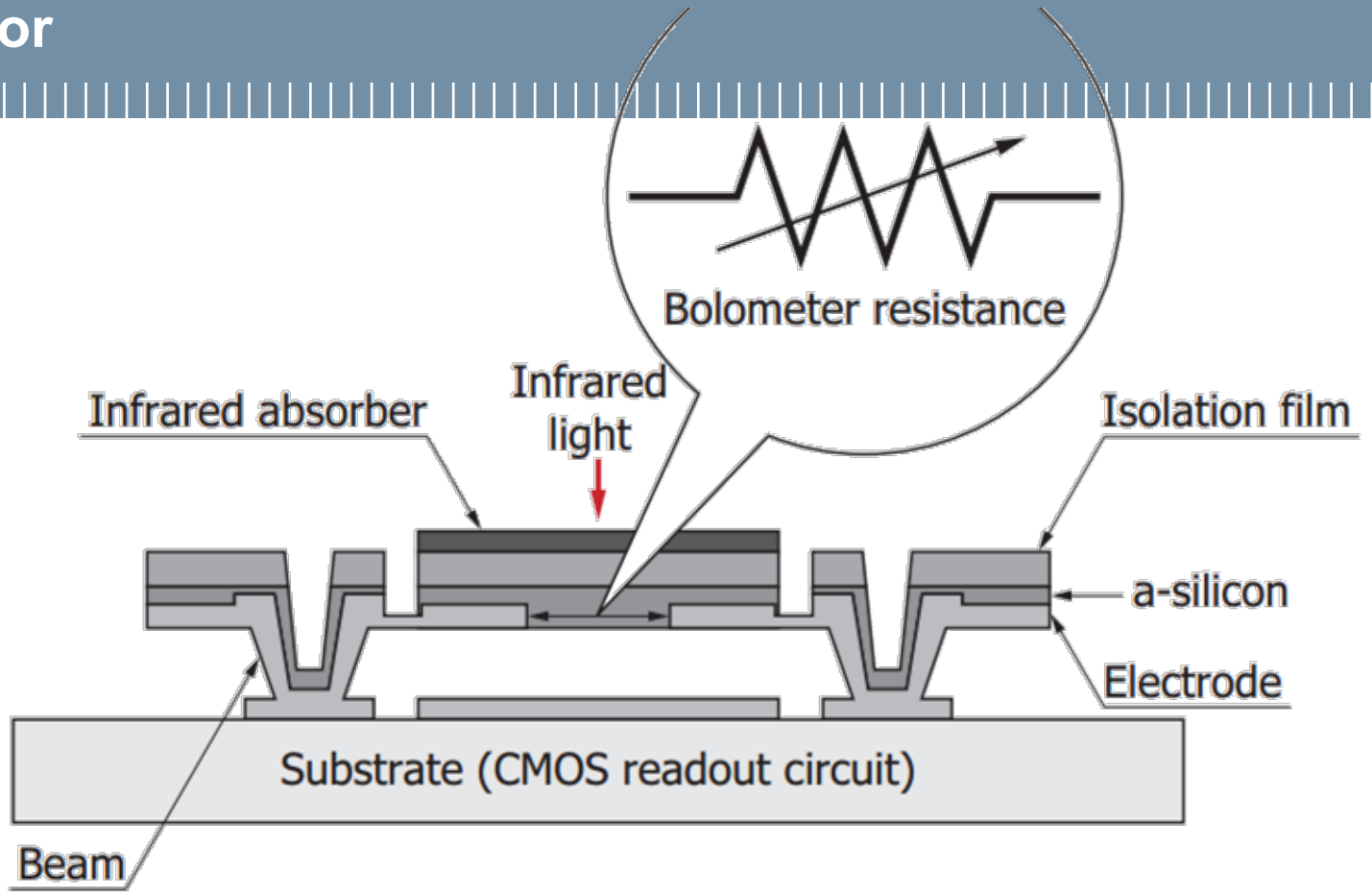


Many thermocouples in series
measure the temperature of the IR absorption film.

Larger photosensitive area → large number of thermocouples → better sensitivity

↓

Typically, single-element detector or array with limited number of elements



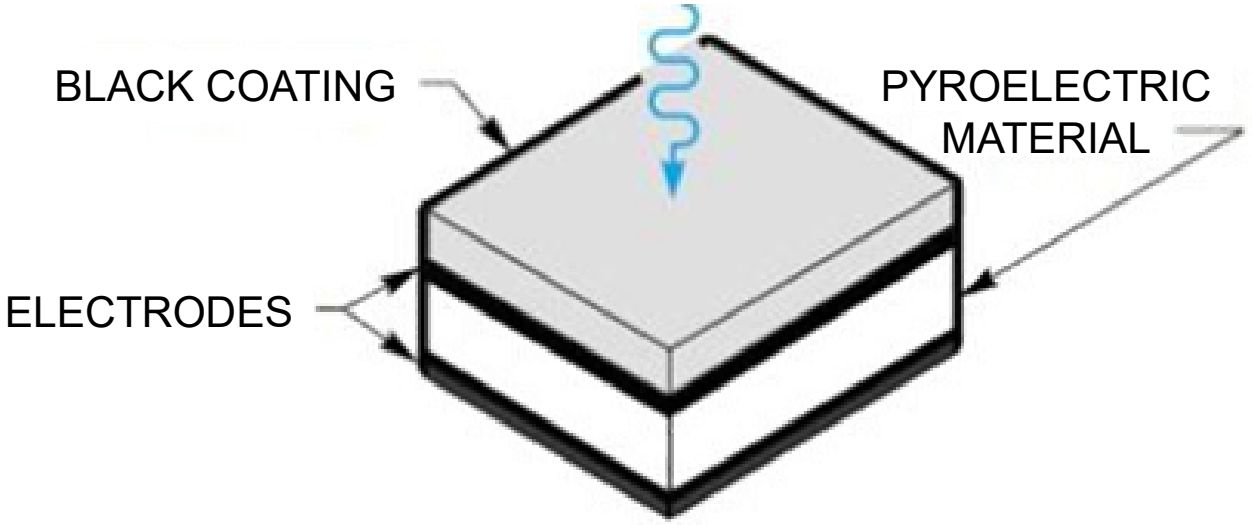
Boolometer resistance = metallic or semiconductor PTC resistor.

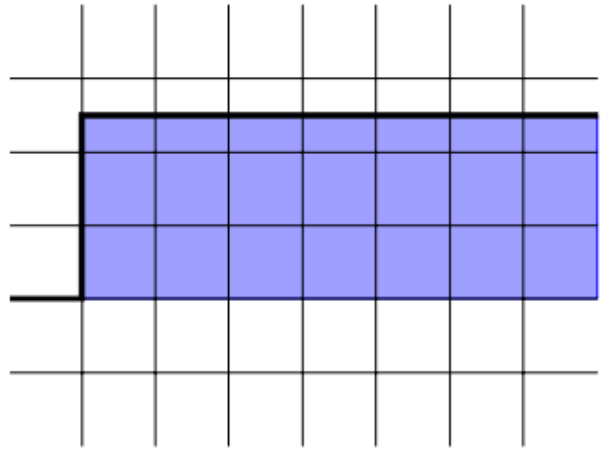
+

Read Out Integrated Circuit to sense the resistance variations.

Thermal isolation between boolometer resistance and CMOS ROIC

Commonly used for multi-pixel cameras.





Infrared input to detector (step)

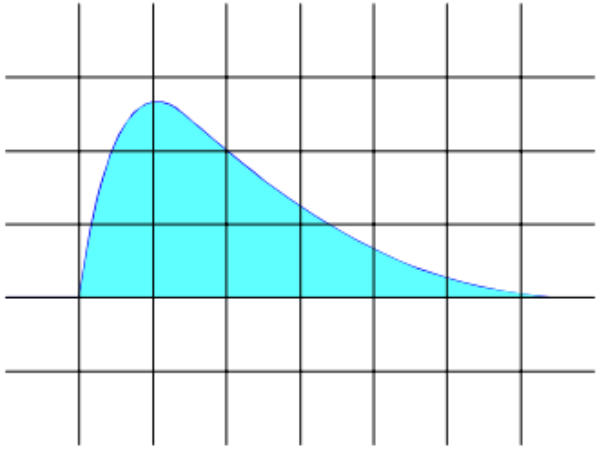
Pyroelectric material: heat variation polarizes dipoles

→ charge is collected on the electrodes
(like in a capacitor)

leakage currents discharge the capacitor

↓

The incoming IR radiation is modulated with optical shutter.



Response of pyroelectric detector

Working principle similar to photodiode (electron-hole pair generation).

- II-VI materials: HgCdTe (Mercury Cadmium Telluride)

  → very low energy gap (0.1-0.4 eV)

- III-V materials: InAs/GaSb (Indium Arsenide-Gallium Antimonide)

  → very low energy gap (0.15 eV) of the heterojunction

Responsivity: $S = \frac{\Delta_{out}}{\Delta_{in}}$

For **thermal detectors** it is independent from λ.
Depends on the efficiency of the absorber and the sensitivity of the temperature sensor.

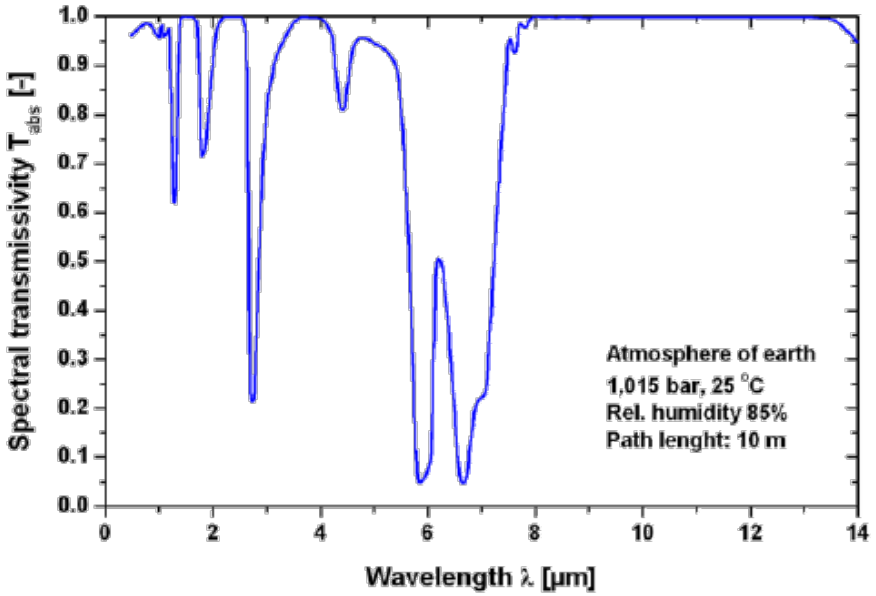
For **quantum detectors** ($S = \frac{I_{out}}{P_{in}}$) it depends on λ, in fact:

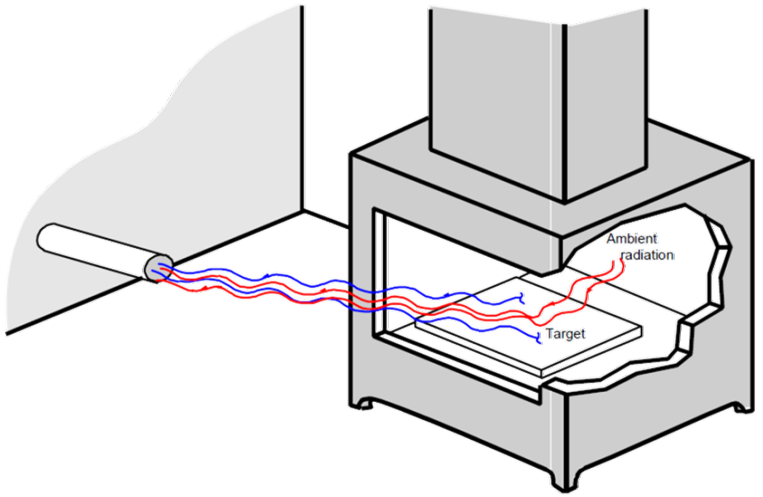
Detector efficiency: $\eta = \frac{n_e}{n_{ph}}$ (n_e electrons generation rate, n_{ph} incoming photons rate)

$$S = \frac{I}{P} = \frac{n_e \cdot q}{n_{ph} \cdot h\nu} = \frac{n_e \cdot q}{n_p \cdot \frac{hc}{\lambda}} = \frac{n_e}{n_p} \cdot \frac{\lambda}{\frac{hc}{q}} = \eta \cdot \frac{\lambda}{1.24}$$

Influence from the surrounding

- Spectral transmissivity of air → wavelength selection
- Ambient radiations → Compensation or shielding
- Dust and particles in the ambient → Auto polishing of lenses





- Maintenance and service in industrial applications

  (moving machines, medium and high voltage facilities…)

- Measuring temperature of electronic circuits
  - Over-heating of electric bus contacts
  - Checking of transformers

- Localization of defective cables



